A certified auxiliary response after shell contact

Ricardo Maldonado
Independent researcher, Henderson, Nevada, USA

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Abstract

Within a retained conditional Einstein–scalar shell construction, we enclose one probability-sphere mean of the auxiliary forced increment at the scaled point $(x, r) = (3/2, 1/4)$. Its backward history reaches the shell after first contact. Exact clipped-area integration of actual-source interval enclosures, combined with a complete coupled mean-block feedback bound, gives

$$55{,}843{,}882 < \mathrm{mean}_{S^3} W_{yy}(3/2, 1/4) < 737{,}593{,}144{,}521.$$

This is a conditional pointwise sign certificate for an auxiliary summand in a registered frame and gauge. It is not a full numerical waveform, an invariant force, or a separate physical solution. The proof uses the inherited smooth first fields, source certificates and linear response construction.

1 Object, inherited framework and result

The retained construction [1, 2] uses factorial second coefficients $g = g_0 + \epsilon q + \epsilon^2 k/2$ and $\phi = \phi_0 + \epsilon\eta + \epsilon^2 w/2$. In the radially parallel frame, the second state $U = (k, w)$ satisfies the complete reduced equation

$$U_{xx} - U_{rr} = AU_x + BU_r + D_0 U + F, \qquad P_D U|_{r=0} = 0, \quad P_N U_r|_{r=0} = MP_N U + G.$$

The scaled coordinates obey $t = \ell x$, $z = \ell r$, $dy = -\rho\, dz$, with $\ell = 10^{-6}$. The exact integral operator is $TV = K(AV_x + BV_r + D_0 V) + L(MP_N V|_{r=0})$. The auxiliary decomposition is

$$U = U_{\mathrm{hom}} + W, \qquad U_{\mathrm{hom}} = (I - T)^{-1} U_{\mathrm{free}}, \qquad W = (I - T)^{-1}(KF + LG).$$

Thus W has zero Cauchy data and retains the complete reduced propagation and Robin feedback. The inherited physical second initial data remain in U_{hom}; zeroing the explicit sources does not produce another physical shell solution. Neither summand is separately claimed to satisfy the full sourced physical constraints.

At $a = 3/2$, $b = 1/4$, let $\lambda W = \mathrm{mean}_{S^3} W_{yy}$, using probability sphere measure and the covariant yy frame component. The certified outward decimal interval is

$$\lambda W(a, b) \in [55{,}843{,}882.796267,\ 737{,}593{,}144{,}520.766258]. \tag{1}$$

All numerical decisions use exact rational fractions. The result remains conditional on the accepted smooth background and actual first master/gauge solutions, their continuous error certificates, and the complete retained second-response operator. These inherited proofs are inputs, not newly established nonlinear existence results.

2 Shell-reaching integral and conventions

Put $d = a - b = 5/4$ and define the physical cones

$$D_+ = \{0 \le s \le a,\ u \ge 0,\ |u-b| \le a-s\}, \qquad D_- = \{0 \le s \le d,\ 0 \le u \le d-s\}.$$

The triangle inequality proves backward closure of D_+, including all reflected dependencies and boundary histories $0 \le s \le d$. Every dependency has $s+u \le a+b = 7/4 < 8$, within the controlled collar. With $J = P_N - P_D$, folding the reflected wave kernel gives

$$Kf(a,b) = \tfrac{1}{2}\int_{D_+} f + \tfrac{1}{2}\int_{D_-} Jf = \tfrac{1}{2}\int_{D_+\setminus D_-} f + \int_{D_-} P_N f,$$
$$Lg(a,b) = -\int_0^d g(s)\,ds.$$

The yy row is Robin, so its two area weights are 1/2 and 1. The negative sign in L is fixed by $\partial_r Lg(x,0) = g(x)$. The Robin matrix remains in T.

The source conversion uses the one-copy outward affine shell Hessian QK, not its doubled two-copy value or an already converted physical forcing. The normal harmonic source and its mean are

$$GN = -2\,\mathrm{tr}_\eta\, QK - 2QC_y, \qquad G_{yy} = 2\ell\rho(-QK_{00} + 3QK_{ab} + QC_y).$$

The time trace is Lorentzian and three is the tangent dimension of S^3. Here QC_y is the full varying-metric-lowered gauge Hessian. Thus

$$\lambda LG = -2\ell\rho_b \int_0^d (-QK_{00,0} + 3QK_{ab,0} + QC_{y,0})\,ds.$$

Here $\rho_b = \rho|_{r=0}$ is time independent. The retained exact zero wedge $s+u<1$ makes the boundary source vanish through time 1. Normal differentiation of shell-eliminated identities is not used.

The selected $n = 16$ partition clips rational polygons at the cone, support and characteristic boundaries. Exact polygon areas multiply source intervals valid on each entire enclosing rectangle; exact boundary lengths multiply whole-interval shell enclosures. Actual master/gauge uncertainties and outward arithmetic are included once in those source ranges. Exterior error improvements are used only when $u_{\min} \ge s_{\max}$ for the whole box. This is interval integration, not point quadrature. The resulting complete source integral $S = KF + LG$ obeys

$$\lambda S(a,b) \in [587{,}660{,}227.409174,\ 737{,}061{,}328{,}176.153350]. \tag{2}$$

3 Exact five-mean closure and coupled feedback

SO(4) averaging of tensor sections commutes with the full equivariant linear bulk and shell operators. Its invariant image has five amplitudes

$$z = (K_0, L_0, N_0, J_0, w_0),$$

corresponding to yy, y0, 00, the sphere-metric coefficient and the scalar. There are no invariant sphere vectors or trace-free sphere tensors. Consequently complementary angular modes cannot linearly feed this block. Every one of the 25 retained primitive matrices independently has zero cross-block entries in both directions. The positive Gram block is $\mathrm{diag}(1,2,1,3,1)$; only L_0 is Dirichlet. Quadratic mean sources produced by nonzero first harmonics are still included before

projection. This closure therefore retains every possible contribution to the mean; it does not assume a scalar-only response.

Let $\overline{A}, \overline{B}, \overline{D}, \overline{M}$ be entrywise absolute majorants of the complete five-by-five operators. They are formed from the exact tensor actions and the inherited background bounds

$$\rho \le 131, \quad |a_{\mathrm{bg}}| \le 14/125, \quad |p| \le 7/50000, \quad |\tanh t| \le 3 \cdot 10^{-6},$$

together with the retained curvature and potential bounds. Angular derivative primitives vanish on the invariant block; connection, curvature, metric/scalar and Robin terms remain. The detailed proof and checker list every term, retaining exact cancellations between primitives with a common scalar coefficient [4].

For each scalar component put

$$X_i = \max\{\sup |z_i|/3,\ \sup |\partial_x z_i|, \sup |\partial_r z_i|\}.$$

These are individual component norms, not H norms; no Frobenius Gram conversion is applied. Symmetric tensor entries are handled by the exact tensor actions. On D_+ the kernel norms satisfy $\|K\| \le a = 3/2$ and $\|L\| \le 1$. Using the retained exact component source caps F_i, G_i, define

$$\Sigma = \overline{A} + \overline{B} + 3\overline{D}, \qquad C = a\Sigma + 3\overline{M}, \qquad f = aF + G.$$

The actual solution obeys $X \le f + CX$ componentwise. All entries of C are nonnegative and its largest row sum is below $0.000853 < 1$. Therefore its nonnegative Neumann inverse gives

$$X \le \beta = (I - C)^{-1} f.$$

The five-dimensional inverse is solved exactly and the identity $\beta = f + C\beta$ is checked. Independent review reconstructs the geometric matrices directly on symmetric covariant tensors and obtains the same resolvent with Cramer determinants.

The exact zero wedge removes a triangle of area $1/2$ from each unfused cone. Since $|D_+| = 47/32$ and $|D_-| = 25/32$, the remaining Robin kernel weight is $5/8$. The actual C^1 trace satisfies $z_i(1, 0) = 0$ and hence $|z_i(s, 0)| \le (s - 1)\beta_i$ for $1 \le s \le 5/4$. It follows that

$$|\lambda(W - S)(a, b)| \le E_5, \qquad E_5 = \tfrac{5}{8}(\Sigma\beta)_K + \tfrac{1}{32}(\overline{M}\beta)_K.$$

The two terms and their complete sum have certified upper displays

Contribution	Upper bound
Bulk feedback	509,975,007.954097
Robin trace feedback	21,841,336.658812
Complete feedback	531,816,344.612908

Subtracting this exact allowance from the exact lower endpoint in (2) gives (1). No extra factor three or angular point-evaluation factor is required: the output is already the scalar mean K_0. Feedback is bounded analytically, not numerically evaluated.

4 Verification and scope

The source partition and continuous source evaluation, the invariant-block majorants, and the final display arithmetic have separate checkers. Input hashes bind the actual source and response reports.

Normal and optimized Python runs agree byte-for-byte. The display generator authenticates its reports before generating the displayed source-integral, feedback, contraction and final response bounds; decimal endpoints use directed rational rounding. Internal independent AI-assisted reviews verify the source ledger, tensor primitives, whole-collar source caps and rational inverse. This is not a claim of external expert review.

The result concerns one point only. No continuous positive region follows without a separate variation bound. No full numerical waveform has been computed. The retained framework already contains a conditional analytic series for total U and its physical constraint completion; the present calculation neither replaces that construction nor repairs a purportedly missing boundary law. The sign of this registered-frame auxiliary mean is not asserted to be invariant under gauge changes, and it supplies no nonlinear family, finite-amplitude remainder, cosmological-origin evidence or general new theorem.

References

[1] R. Maldonado, *Full second coefficient on the retained causal collar*, conditional retained construction, 16 October 2026. Bundled as `FULL_SECOND_RESPONSE_CONSTRUCTION.md`.

[2] R. Maldonado, *Separating the forced increment in the retained formal second response*, retained conditional corollary. Bundled as `FORCED_INCREMENT_BOUND.md`.

[3] *Continuous actual evolved geometric source cell enclosures*, retained exact source report and source dependency, 2026. `ACTUAL_SOURCES_NORMAL.json`.

[4] *Conditional positive mean at the shell-reaching point*, detailed proof, exact certificate and independent review, 23 October 2026. Accompanying reproducibility files.